



BRAC UNIVERSITY

Dept. of Computer Science and Engineering

CSE250L

Circuits and Electronics Laboratory

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Experiment No. 4

Study of I-V Characteristics of Linear Circuits

Objective

This experiment aims to acquaint students with the concept of I-V characteristics. They will find I-V characteristics of some linear components and some circuits consisting linear combinations of them.

Part 1: By Using Multimeter

Theory

I-V characteristics, also known as current-voltage characteristics, describe the relationship between the current flowing through a device/circuitry and the corresponding potential difference (voltage) across it. This concept is commonly used in the field of electronics and electrical engineering to analyze the behavior of various components such as resistors, diodes, transistors, and, in general, circuits.

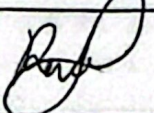
I-V characteristics provide a way to understand how current and voltage interact in electrical and electronic components and circuits. By analyzing these characteristics, circuits/devices can be designed and optimized, appropriate components can be selected, and the behavior of devices under different operating conditions can be predicted.

In hardware labs, studying the I-V characteristics of an element/circuitry can be done in some simple steps. After building the circuit using hardware tools (such as Breadboards, Power Supply), a multimeter or other measuring instruments can be used to measure the voltage and current at specified terminals in the circuit. The multimeter probes can be placed across the component or along the desired path (specified by the terminals) to measure the voltage difference and current flow. To determine the I-V characteristics, the voltage or current across the circuit or specific components must be varied. This can be done by adjusting the power supply voltage, using variable resistors, or changing the values of other circuit parameters. As the voltage or current is varied and the corresponding values are measured, the data can be recorded in a table. The voltage and current values for each point of interest in the circuit should be noted.

Once we have the values of currents and voltages at various points in the circuit, the I-V characteristics can be plotted. Typically, this involves creating a graph with current (I) on the y-axis and voltage (V) on the x-axis.

Data Tables

Signature of Lab Faculty:



Date:

8/14/25

**** For all the data tables, take data up to three decimal places, round to two, and then enter into the table.**

Table 0: Resistance Data

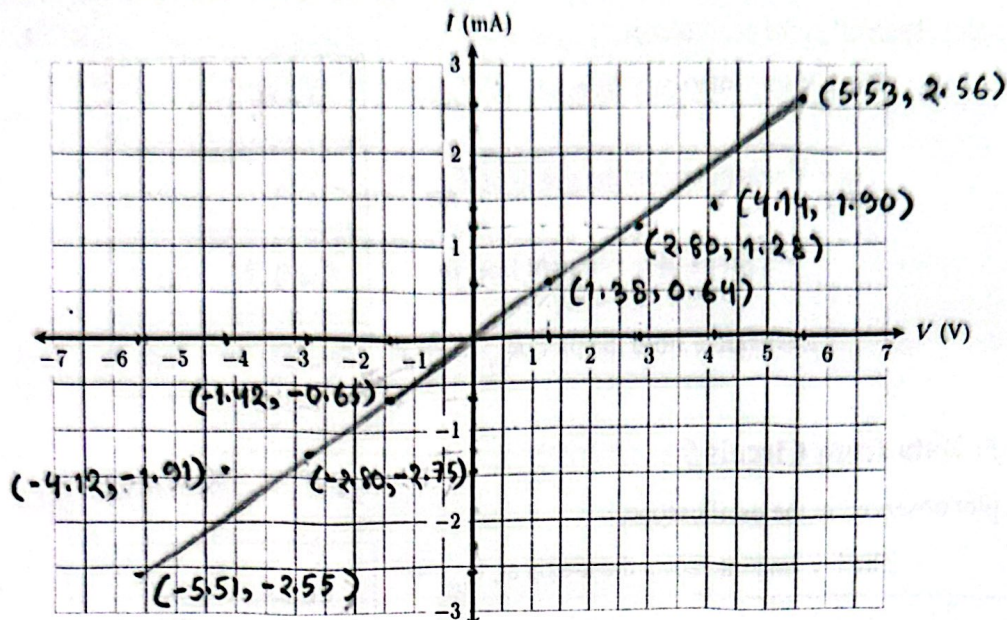
For all your future calculations, please use the observed values only (even for theoretical calculations).

Notation	Expected Resistance	Observed Resistance (k Ω)	Notation	Expected Resistance	Observed Resistance (k Ω)
R_s	1 k Ω	0.984	R_3	3.3 k Ω	3.256
R_1	1 k Ω	0.988	R_4	4.7 k Ω	4.63
R_2	2.2 k Ω	2.166	R_5	10 k Ω	10.08

Table 1: Data from Circuit 1

V_s (V)			V_1 (V)		V (V)		$I = \frac{V_1}{R_1}$ (mA)	
Expected Voltage	From DC power supply	Using multi-meter	Experimental	Theoretical	Experimental	Theoretical	Experimental	Theoretical
-8.0	-8.0	-8.06	-2.51	-2.5	-5.51	-5.5	-2.55	-2.5
-6.0	-6.0	-6.0	-1.88	-1.875	-4.12	-4.125	-1.91	-1.875
-4.0	-4.0	-4.05	-1.27	-1.25	-2.80	-2.75	-1.29	-1.25
-2.0	-2.0	-2.03	-0.64	-0.625	-1.42	-1.375	-0.65	-0.625
0.0	0.0	-0.003	-0.001	0	-0.002	0	-0.00101	0
2.0	2.0	2.03	0.63	0.625	1.38	1.375	0.64	0.625
4.0	4.0	4.05	1.26	1.25	2.80	2.75	1.28	1.25
6.0	6.0	6.0	1.87	1.875	4.14	4.125	1.90	1.875
8.0	8.0	8.06	2.52	2.5	5.53	5.5	2.56	2.5

Plot the values of I and V from the above table.



Draw the best-fitting straight line through all the data points.

Slope of the straight line, $m =$

0.46 $\text{k}\Omega^{-1}$

Resistance from the plot, $R_{eq} = \frac{1}{m} =$

2.17 $\text{k}\Omega$

Percentage of Error = $\left| \frac{R_2 - R_{eq}}{R_2} \right| \times 100\% =$

0.185 %

Table 2: Data from Circuit 2

For the plot observed in the oscilloscope,

Does the I-V line pass through the origin? ($\checkmark$ / $\times$)



Slope of the straight line, $m =$

0.484 $\text{k}\Omega^{-1}$

Resistance from the plot, $R_{eq} = \frac{1}{m} =$

2.066 $\text{k}\Omega$

Table 3: Data from Circuit 3

For the plot observed in the oscilloscope,

Is the I-V line parallel to y-axis? ($\checkmark$ / $\times$)



The straight line intersects the x-axis at, $V_{oc} =$

34 V

Table 4: Data from Circuit 4

For the plot observed in the oscilloscope,

The I-V line intersects the x-axis at, $V_{oc} =$	3.4	V
The I-V line intersects the y-axis at, $I_{sc} =$	-1.6	mA
Slope of the straight line, $m =$	0.47	$k\Omega^{-1}$
Resistance from the plot, $R_{eq} = \frac{1}{m} =$	2.13	$k\Omega$

Table 5: Data from Circuit 5

For the plot observed in the oscilloscope,

The I-V line intersects the x-axis at, $V_{oc} =$		V
The I-V line intersects the y-axis at, $I_{sc} =$		mA
Slope of the straight line, $m =$		$k\Omega^{-1}$
Resistance from the plot, $R_{eq} = \frac{1}{m} =$		$k\Omega$

Table 6: Observation from Circuits 6

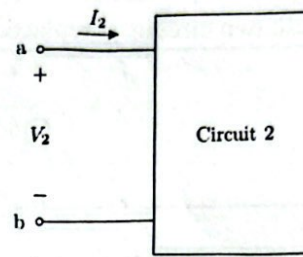
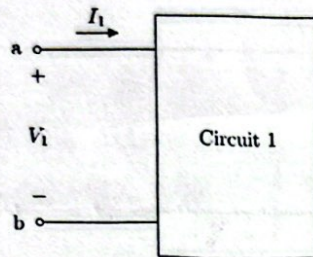
Potentiometer fully CW $\rightarrow$ I-V line becomes (x-axis or y-axis) -	x-axis
Potentiometer fully CCW $\rightarrow$ I-V line becomes (x-axis or y-axis) -	y-axis

Table 7: Observation from Circuits 7

LDR is in darkness $\rightarrow$ its resistance ($\uparrow$ or $\downarrow$) -	$\uparrow$
LDR is illuminated $\rightarrow$ its resistance ($\uparrow$ or $\downarrow$) -	$\downarrow$

Questions

1. What conditions must exist for the following two circuits to be equivalent to each other with respect to terminals $a - b$?



Condition 1: $I_1 = I_2$

Condition 2: $V_1 = V_2$

2.

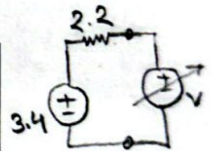
- (a) For Circuit 4, you constructed it in the laboratory, derive a relation between I and V .

[Hint: Exclude the experimental setup part (left side of V) in Circuit 4. Your equation cannot contain any variables other than I and V]

Excluding left side of V

$$\Rightarrow I = \frac{1}{2.2} V + \left(\frac{-3.4}{2.2} \right)$$

KVL, $-3.4 - 2.2I + V = 0$



- (b) For Circuit 5, you constructed it in the laboratory, derive a relation between I and V .

[Hint: The $10\text{ k}\Omega$ resistor can be ignored as it is parallel to a voltage source. Exclude the experimental setup part (left side of V) in Circuit 5 and try to reduce the circuit to a single loop first. Your equation cannot contain any variables other than I and V]

Excluding left side of V and $10\text{ k}\Omega$

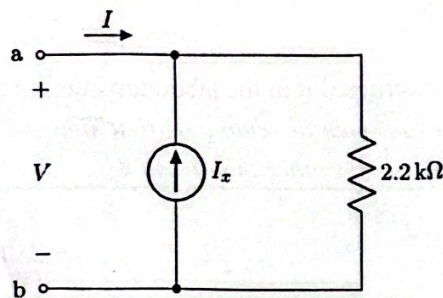
(c) Do the $I - V$ equations derived in (a) and (b) closely match?

☐ Yes ☐ No

If yes, what conclusion can you draw from this, and will it have any effect if one of these two circuits is replaced with the other? Why??

(d) Now, for the following circuit, determine the value of I_x so that the $I - V$ relation matches with those you derived in (a) or (b). Is this circuit equivalent to Circuit 4 and Circuit 5?

[Hint: Derive a relation between I and V from this circuit. Then compare it with that in (a) or (b) to get I_x .]



KCL, Current through $2.2\text{ k}\Omega$
 $= I + I_x$

So, $V = (I + I_x) 2.2$

$$I = \frac{1}{2.2} V + (-I_x) \quad \text{--- (1)}$$

Comparing with a,

From 'a', $I = \frac{1}{2.2} V + \frac{-3.4}{2.2} \quad \text{--- (1)}$

$$\therefore I_x = \frac{3.4}{2.2}$$

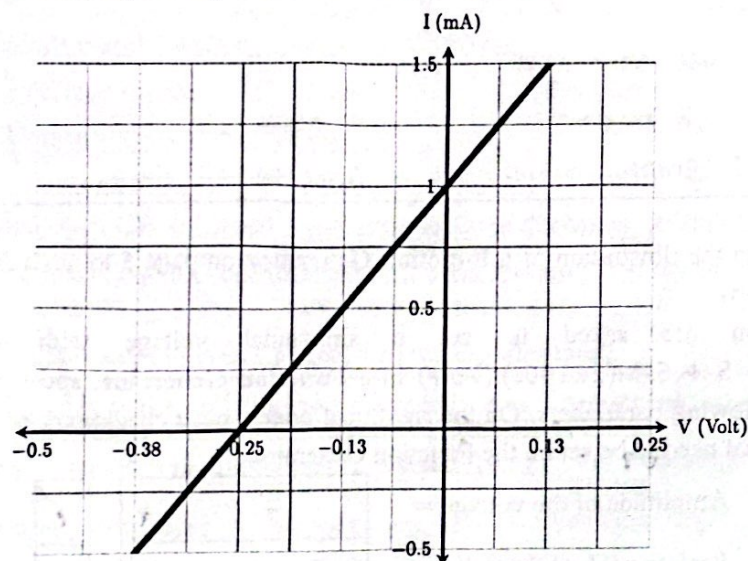
Yes, this circuit is equivalent to Circuit 4.

3. How can you directly measure the voltage-axis and current-axis intersecting points of the $I - V$ line of any linear circuit?

For voltage axis,
let, $I = 0$
then solve for V .
 V is the intercept
of voltage axis.

For current axis,
let, $V = 0$
then solve for I
 I is the intercept
of Current axis.

4. A linear two-terminal circuit has the following $I - V$ relationship at the terminals $a - b$ measured in a laboratory with the setup shown below. Draw (inside the dashed box) a simplified version of the circuit that can give rise to the same $I - V$ as shown. Also, write the corresponding values of the circuit elements.



$$IR_a + V_a + V = 0$$

$$\Rightarrow I = \frac{1}{R_a}V + \left(\frac{V_a}{R_a}\right)$$

$$m = \frac{\Delta V}{\Delta I} \quad R_a = \frac{\Delta I}{\Delta V}$$

X axis,
 $I = 0$

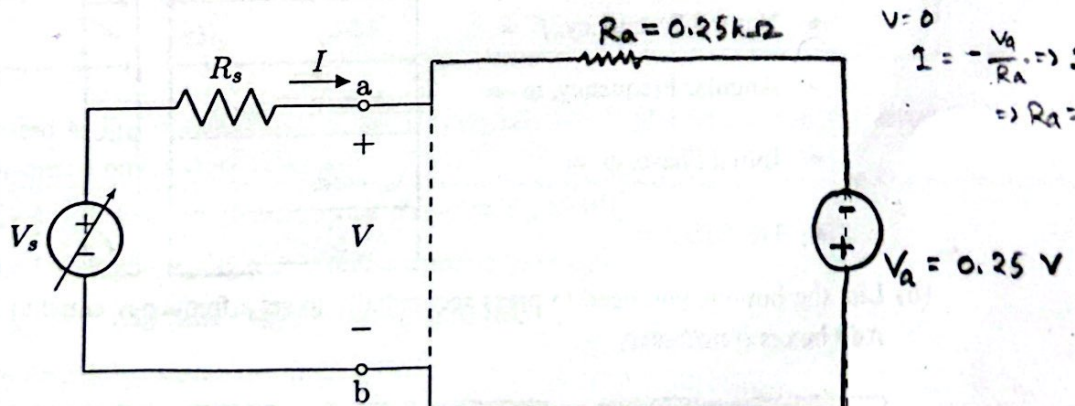
$$V = V_a = -0.25$$

Y axis,

$$V = 0$$

$$I = -\frac{V_a}{R_a} \Rightarrow I = -\frac{-0.25}{R_a}$$

$$\Rightarrow R_a = 0.25$$



Draw inside the dashed box

$$I = \frac{1}{0.25}V + \frac{-0.25}{0.25}$$

5. Refer to the illustration of an Oscilloscope on page 5 to answer the following questions—

- (a) In the normal mode of operation, an oscilloscope always plots—
☒ voltage as a function of time ☐ current as a function of time
- (b) In the X-Y mode of operation, an oscilloscope plots Channel-1 along the -
☐ x-axis ☒ y-axis
- (c) What is the function of the button “Measure”?
☐ It measures the total resistance of the circuit.
☐ It adds the voltage waveforms fed to the channels.
☒ It displays the properties of the voltage waveforms fed to the channels.
- (d) Is there any way to observe the one you haven't selected in (a)?

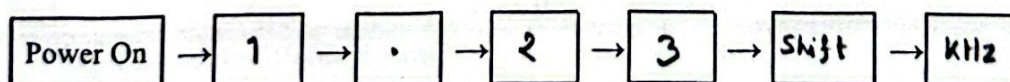
In x-y mode oscilloscope shows us V vs V graph.
 We know, $V = IR \Rightarrow I = \frac{V}{R}$ [Ohm's law]
 If we use very small R then
 $I = V$. In this way the V vs V graph will technically show us
 the I graph, current as a function of time.

6. Refer to the illustration of a Function Generator on page 5 to answer the following questions:

- (a) If you are asked to set a sinusoidal voltage with a dc offset $v(t) = 5 + 5\sin(2\pi 100t)$ (Volt) in a Function Generator, specify the values of the following parameters. On the rightmost boxes, put a checkmark ✓ to indicate the ones that need to be set on the Function Generator.

• Amplitude of the voltage =	5 V	☒
• Peak to peak of the voltage =	$2 \times 5 = 10$ V	☐
• Natural Frequency, f =	100 Hz	☒
• Angular Frequency, ω =	200π rads^{-1}	☐
• Initial Phase, ϕ =	0 °	☐
• DC Offset =	5 V	☒

- (b) List the buttons you need to press sequentially to set a frequency equal to 1.23 kHz.
 Add boxes if necessary.



(c) Turning the "AMPL" knob clockwise or counterclockwise will change the—

- ☐ Frequency of the voltage waveform
- ☒ Amplitude of the voltage waveform
- ☐ Phase of the voltage waveform

(d) What is the function of the button "WAVE"?

- ☐ It converts a voltage waveform into a current waveform
- ☐ It sets everything to default.
- ☒ It changes the types of voltage waveforms.

7. Put checkmarks (✓) to indicate the correct answers. The questions may have more than one answer.

(a) The I-V characteristics of which circuits went **through origin** -

- | | | |
|---|---|---|
| ☒ Circuit 2 | ☐ Circuit 3 | ☐ Circuit 4 |
| ☐ Circuit 5 | ☒ Circuit 6 | ☒ Circuit 7 |

(b) Circuits 6 and 7 were **equivalent to a resistor** -

- | | | |
|---|--|-----------------------------------|
| ☐ Voltage source | ☒ Current source | ☐ Resistor |
| ☐ Short Circuit | ☐ Open source | |

(c) When the LDR in Circuit 7 was completely **in darkness**, it was equivalent to -

- | | | | |
|--|--|--|---|
| ☐ short circuit | ☒ open-circuit | ☐ 1 k Ω resistor | ☒ 0A current source |
|--|--|--|---|

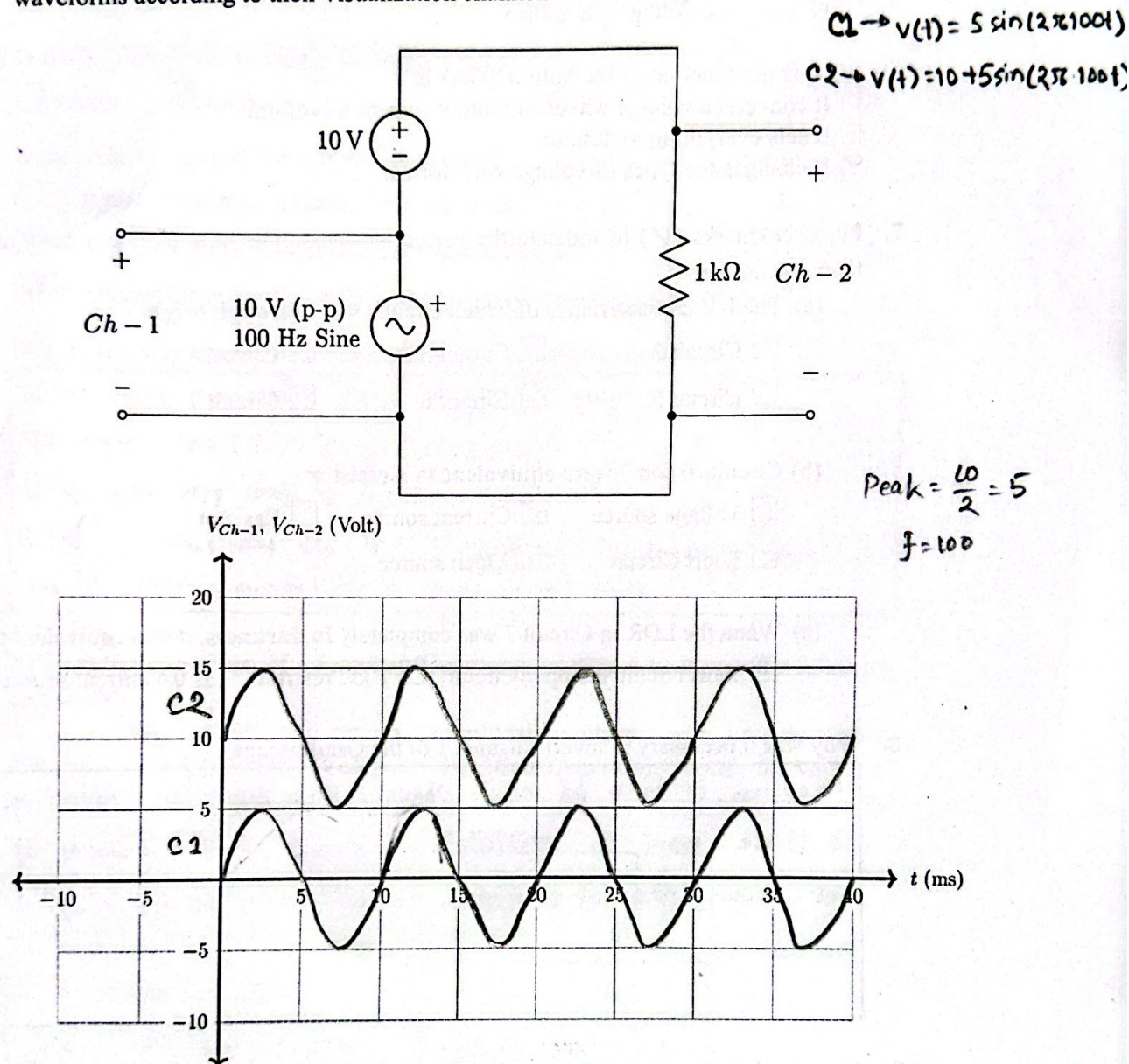
8. Why was it necessary to invert Channel 1 of the Oscilloscope?

Negative terminal of each channel are connected internally. So if we want to separate the pole in requirement of our circuit, channel 1 needs to be inverted to give actual result.

9. Compared to the expected one, what the observed $I - V$ plot be if we didn't invert the Channel 1 on the Oscilloscope?

- ☐ The $I - V$ plot would mirror with respect to the y-axis.
- ☒ The $I - V$ plot would mirror with respect to the x-axis.
- ☐ The $I - V$ plot would still be the expected one.
- ☐ We wouldn't observe any graph.

10. Approximately draw the waveforms that should be observed in Channel-1 and Channel-2 of an oscilloscope when both the channels are ON and are connected in the setup shown below. Draw both plots in the same template given below. Mark the waveforms according to their visualization channel.



Report

1. Fill up the theoretical parts of all the data tables.
2. Answers to the questions.
3. Attach the captured images of the $I - V$ plot observed for Circuits 2, 3, 4, 5, 6, and 7 in the Oscilloscope. Fit all the images on a page or two and print.
4. Discussion [your overall experience, accuracy of the measured data, difficulties experienced, and your thoughts on those]. Add pages if necessary.

This lab was a new experience to me. For the first time I encountered oscilloscope in this lab. For me this lab was a bit complex. As I were new to those device at first It was confusing for me to take readings. But as I with my group continued working we kept figuring out. Also our data were quite accurate.

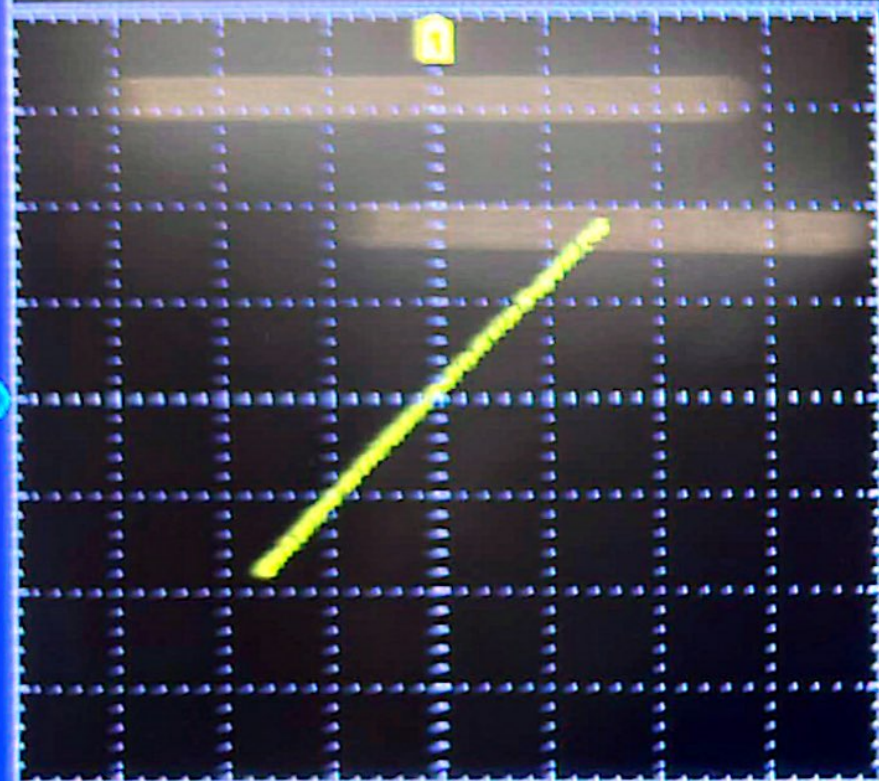
GW17570K

10k pts

2MSa/s



Trig'd



① 5.00V

② 2.00V

500us

0.00000s

F 500.000Hz

② f -80mV DC

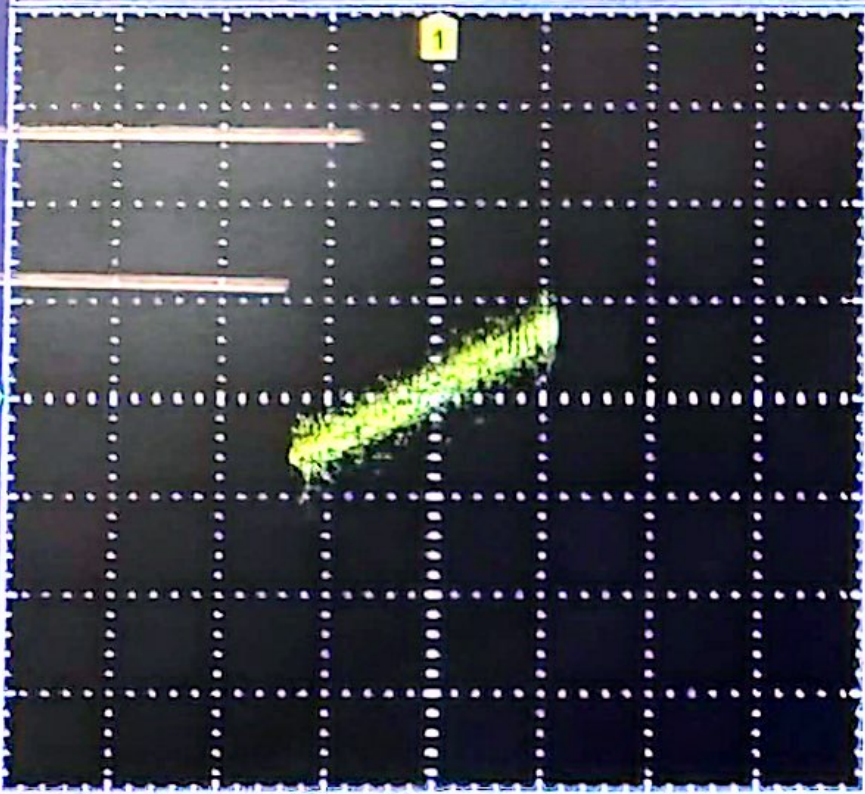
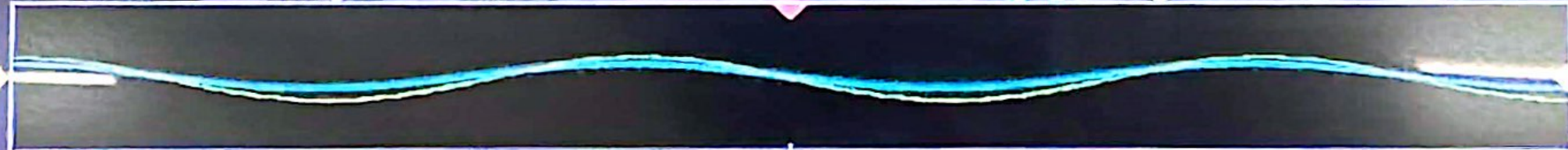
GWINSTEK

10k pts

2MS/s



Trig'd



① ↓ = 10.0V ② = 200mV

500ns

0.00000s

F 499.996Hz

① f 0V DC

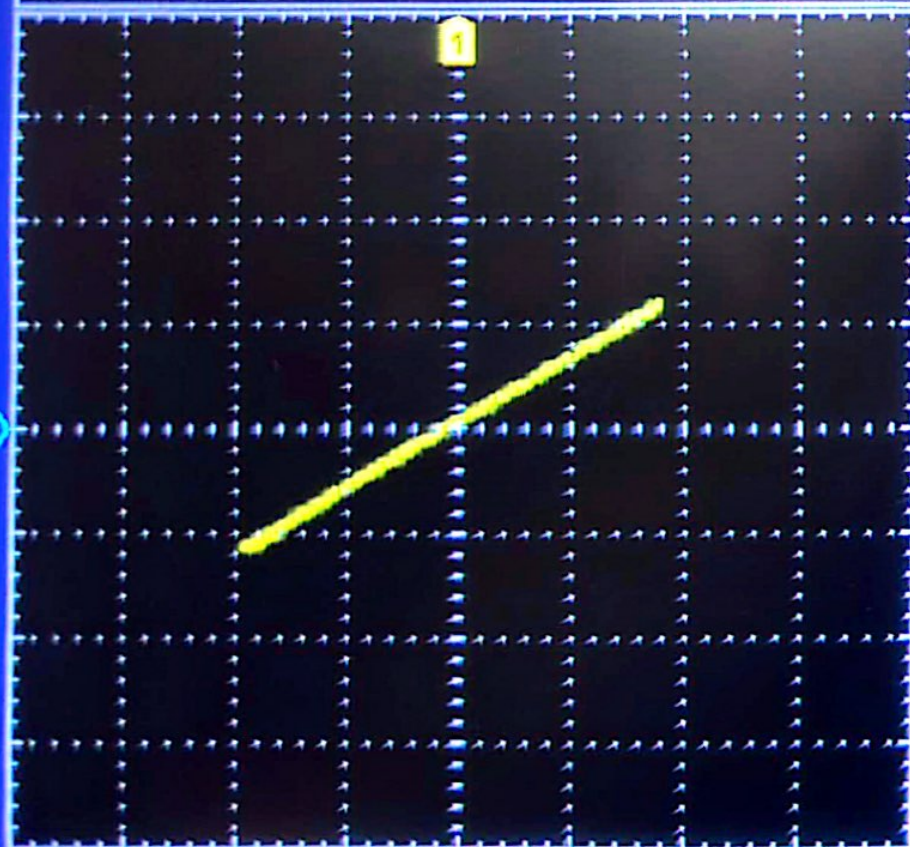
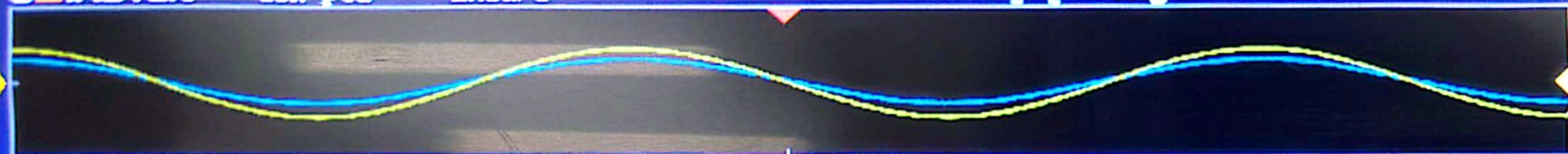
GWINSTEK

10k pts

2MSa/s



Trig'd



① ↓ = 5.00V

② = 2.00V

500us

0.00000s

F 500.005Hz

①

f

0V

DC

GWINSTEK

10k pts

2MSa/s



Trig'd

2

2

1

Vertical Position

F 500.001Hz

1 ↓ = 1.00V 2 = 10.0V

500us 0.00000s

2 f -3.6V DC

2

Coupling
DC AC GND

Impedance
1MΩ

Invert
On Off

Bandwidth
Full

Expand
By Ground

Position /
Set to 0
0.000V

Probe
Voltage
1 X

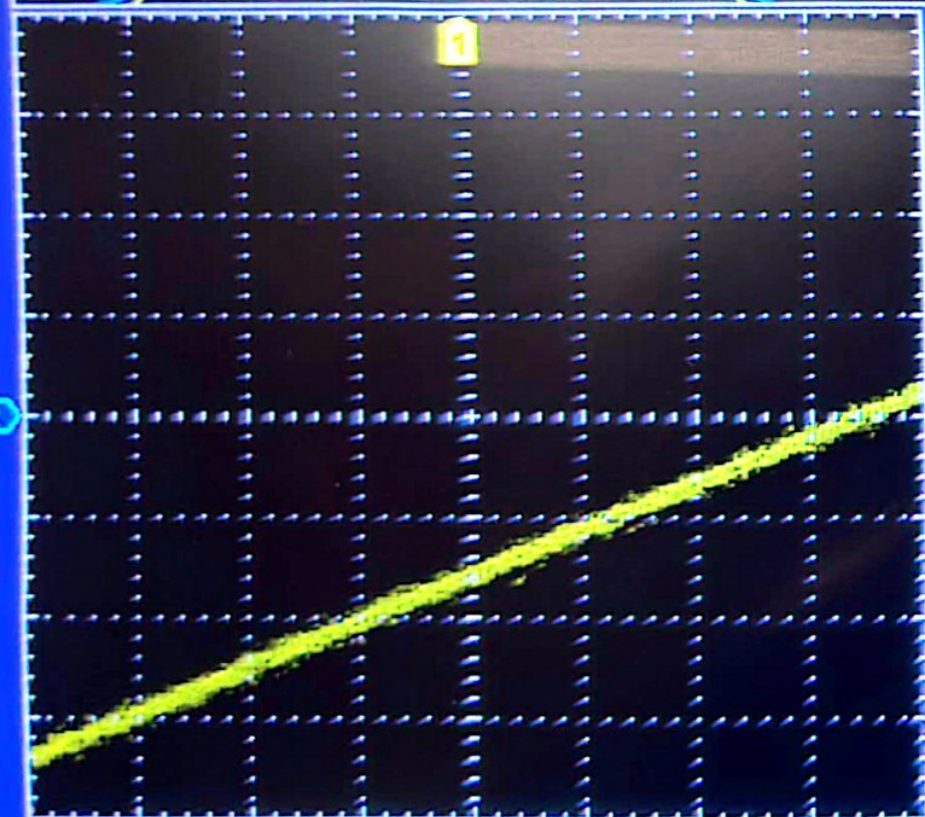
GWINSTEK

18k pts

2MSa/s



Trig'd



① $\downarrow = 1.00V$ ② $\uparrow = 1.00V$

500us 0.00000s

F 499.985Hz

② $\uparrow -1.12V$ DC